Fluency with the four operations

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Calculate 638,475 + 274,806 and 902,140 - 386,759. Copy or complete the first step.

2. Calculate 6,304 x 27.

3. Calculate 8,736 / 24.

4. Miro says: Miro switches methods without checking which operation the question uses. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Calculate 638,475 + 274,806 and 902,140 - 386,759. Copy or complete the first step.
   Answer 913,281 and 515,381.

2. Calculate 6,304 x 27.
   Answer 170,208.

3. Calculate 8,736 / 24.
   Answer 364.

4. Miro says: Miro switches methods without checking which operation the question uses. Is this correct? Explain.
   Answer Mark the operation and estimate before beginning the exact method.

Fluency with the four operations

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: ___________________

1. Calculate 638,475 + 274,806 and 902,140 - 386,759.

2. Calculate 6,304 x 27.

3. Calculate 8,736 / 24.

4. Solve this independently and show every stage: Calculate 638,475 + 274,806 and 902,140 - 386,759.

5. Use a second method or representation for: Calculate 6,304 x 27.

6. Check this result using an inverse, estimate or known fact: Calculate 8,736 / 24.

7. Choose operations to make a true equation using 48, 12 and 4.

8. Identify and correct this misconception: Miro switches methods without checking which operation the question uses.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Calculate 8,736 / 24.

Teacher answers and checking prompts

1. Calculate 638,475 + 274,806 and 902,140 - 386,759.
   Answer 913,281 and 515,381.

2. Calculate 6,304 x 27.
   Answer 170,208.

3. Calculate 8,736 / 24.
   Answer 364.

4. Solve this independently and show every stage: Calculate 638,475 + 274,806 and 902,140 - 386,759.
   Answer 913,281 and 515,381.

5. Use a second method or representation for: Calculate 6,304 x 27.
   Answer Expected result: 170,208. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Calculate 8,736 / 24.
   Answer Expected result: 364. A valid check must be shown.

7. Choose operations to make a true equation using 48, 12 and 4.
   Answer Examples include 48 / 12 = 4 and 12 x 4 = 48.

8. Identify and correct this misconception: Miro switches methods without checking which operation the question uses.
   Answer Mark the operation and estimate before beginning the exact method.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Calculate 8,736 / 24.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Fluency with the four operations

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Choose operations to make a true equation using 48, 12 and 4.

2. Prove the answer to this question, rather than only calculating it: Calculate 638,475 + 274,806 and 902,140 - 386,759.

3. Solve this in two different ways and compare them: Calculate 6,304 x 27.

4. Work backwards from the answer to reconstruct the question: 364.

5. Explain why this reasoning fails: Miro switches methods without checking which operation the question uses.

6. Create a new example that tests the same idea as: Calculate 8,736 / 24.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Choose operations to make a true equation using 48, 12 and 4.
   Answer Examples include 48 / 12 = 4 and 12 x 4 = 48.

2. Prove the answer to this question, rather than only calculating it: Calculate 638,475 + 274,806 and 902,140 - 386,759.
   Answer Expected result: 913,281 and 515,381. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Calculate 6,304 x 27.
   Answer Expected result: 170,208. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 364.
   Answer One valid reconstruction is: Calculate 8,736 / 24. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro switches methods without checking which operation the question uses.
   Answer Mark the operation and estimate before beginning the exact method.

6. Create a new example that tests the same idea as: Calculate 8,736 / 24.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.